

Coral Reef Regime Classification

Problem statement

Coral reefs are vital marine ecosystems but face growing stress from human and natural influences (warming, turbidity, storms). This project predicts coral reef regimes — categorical reef health states (Low, Medium, High) — from environmental and reef survey variables to support monitoring and management decisions.

Approach

I framed the task as a supervised multiclass classification problem. From a small compiled reef table (~620 records), I selected numeric predictors (depth, percent cover, bleaching metrics, sea-surface temperature indices, winds, etc.), handled missing values with mean imputation, standardized features, and addressed class imbalance. Models evaluated included:

- Logistic Regression (baseline, GridSearchCV)
- Support Vector Machine (baseline, GridSearchCV, PCA+SVM)
- Random Forest (baseline)

Implementation overview

Data cleaning and preprocessing were executed in dedicated notebooks. Key steps:

- Load CSV, replace non-data tokens ('nd') with NaN, coerce numeric types.
- Drop rows missing core targets (regime classes); impute remaining NaNs with column means.
- Create multiclass target ('regime 1', 'regime2', 'regime3', 'regime5')
- Train/test/validation split (400 samples, 100 samples, 100 samples respectively).
- Scale numeric features with StandardScaler;
- Logistic Regression: baseline; tuned with GridSearchCV(exhaustive) produced the best LR (~78%).
- SVM: baseline RBF, tuned with GridSearchCV, and PCA+SVM (95% variance). PCA gave the best SVM with accuracy of 78%.
- Random Forest: baseline ensemble (n_estimators=200, class_weight='balanced') — best performer at 86% accuracy.

Key results (ranked by test accuracy)

1. Random Forest (baseline) — 86%
2. SVM (GridSearchCV) — 76%
3. SVM (PCA+SVM) — 78%
4. SVM (baseline) — 75%
5. Logistic Regression (GridSearchCV) — 78%
6. Logistic Regression (baseline) — 73%

Conclusions

The Random Forest baseline showed the strongest predictive performance on the assembled dataset, suggesting the target patterns are well-captured by an ensemble of decision trees without heavy tuning. SVM with PCA provided a close second, proving that focused hyperparameter search can yield near-ensemble performance. Logistic Regression — even after tuning — underperformed relative to non-linear methods, indicating non-linear relationships and interactions are important for this dataset.

Challenges

- Missing and mixed-type fields required careful coercion and imputation; many rows contained 'nd' placeholders.
- Hyperparameter searches can be computationally expensive;
- Less number of available data and possibility of overfitting.